Questions by topic: Importance of biodiversity: resources & ecosystem services - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 1 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 1 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

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0 1 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

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0 2 Table 1 shows characteristics introduced into commercial crops from crop wild relatives (CWRs).

Table 1

Crop	Wild relative source	Characteristic introduced
Sugar cane	Wild sugar cane, Indonesia	Resistance to mosaic virus
Tomato	Single wild species	Resistance to nine major diseases
Rice / barley	Wild salt-marsh varieties	Salt tolerance
Oil palm	Wild palms, central Africa	25% higher yield
Wheat	Wild wheat	Better mycorrhizal associations / phosphate uptake

0 2 . 1 Explain why commercial crop varieties often lack the characteristics shown in Table 1.

[3 marks]

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0 2 . 2 Explain what is meant by a **Vavilov centre** (centre of diversity) and why these areas are conservation priorities.

[3 marks]

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0 2 . 3 Explain why a large population does not necessarily have a large gene pool, and why a small gene pool is a problem.

[3 marks]

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0 2 . 4 Give **one** wild plant species with potential as a new commercial food crop and the characteristic that makes it valuable.

[1 mark]

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0 3 . 1 Explain how living organisms provide each of the following ecosystem services.

[6 marks]

Regulation of atmospheric composition .....

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Soil maintenance .....

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Interspecies relationships .....

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0 3 . 2 Explain why Darwin's orchid of Madagascar depends on a single moth species, and why such specialisation is both an advantage and a risk.

[2 marks]

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0 3 . 3 Explain why elephants are particularly important seed dispersers.

[2 marks]

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0 4 . 1 Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

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0 4 . 2 State what is meant by **biodiversity**.

[1 mark]

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0 5 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 5 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 5 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 6 Biomimetics uses knowledge of the adaptations of other species to improve the design of manufactured items.

0 6 . 1 Complete Table 1 by adding the missing adaptation or application.

[4 marks]

Table 1

Species and adaptation	Design application	
Tubercles (bumps) on the flippers of humpback whales		
	Self-cleaning glass	
Shark skin scales that reduce friction		
	Velcro	

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0 6 . 2 Explain how the design of termite mounds has been copied to reduce the environmental impact of large buildings.

[3 marks]

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0 6 . 3 Describe **two** ways in which studying the physiology of other species could help to solve human health problems.

[2 marks]

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0 6 . 4 Explain why bird wing bones are a useful model for engineers.

[1 mark]

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0 7 . 1 Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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0 7 . 2 Explain how soil organisms provide services that humans depend on.

[3 marks]

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0 7 . 3 Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

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0 7 . 4 Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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0 8 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

0 8 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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0 8 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

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0 8 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

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0 8 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

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END OF QUESTIONS